

# LITERATURE STATUS NOTE: COGENERATORS AND WEAK STABILITY

FA\_BANACH\_001, AGENT\_LANE\_03

## STATUS

**Verdict.** The cogenerator weak-stability question extracted from Eisner–Farkas–Nagel–Sereny [1] is **literature\_already\_answered** by O’Brien [2].

This is not a new proof or counterexample from the run. It is a duplicate-memory packet recording that a later paper explicitly identifies and answers the same problem.

## SOURCE QUESTION

The source is arXiv:0805.1039, “Weakly and almost weakly stable  $C_0$ -semigroups” by Tanja Eisner, Balint Farkas, Rainer Nagel, and Andras Sereny [1]. In the closing section “Comments and open questions,” the authors recall their cogenerator theorem for strong stability: if  $(T(t))_{t \geq 0}$  is a  $C_0$ -semigroup of contractions with cogenerator  $G$ , then

$$\lim_{t \rightarrow \infty} \|T(t)x\| = \lim_{n \rightarrow \infty} \|G^n x\|,$$

so strong stability of the semigroup is equivalent to strong stability of  $G$ . Immediately afterward they ask whether the analogue is true for weak stability, more precisely whether there is a connection between weak stability of a contractive semigroup and weak stability of its cogenerator.

In the book version cited by the later paper, this is Open Question 2.23 of T. Eisner’s *Stability of Operators and Operator Semigroups*.

## SUPPORTING ANSWER

The supporting paper is arXiv:2005.03821, “Contractions, Cogenerators, and Weak Stability” by Robert E. O’Brien [2]. Its abstract states that a  $C_0$  contraction semigroup on a Hilbert space and its cogenerator define a limit algebra that gives a necessary and sufficient condition for the two to be weakly stable equivalent. The introduction further says that this condition addresses the open question from Eisner.

The decisive identification appears in the remark following Theorem 5.3. O’Brien states that weak stability of a  $C_0$  contraction semigroup  $\mathcal{T}$  is equivalent to weak stability of its cogenerator semigroup  $\mathcal{D} = \{T^n : n \geq 0\}$  if and only if their unitary parts have a common limit algebra, equivalently are entangled. The same remark explicitly says that this addresses Eisner’s Open Question 2.23.

## SCOPE AND LIMITATIONS

The answer is a criterion, not an unconditional equivalence theorem. It says exactly when weak stability transfers between the Hilbert-space contraction semigroup and its cogenerator: the unitary parts must have a common limit algebra. Thus the extracted cogenerator question should not be pursued as a new open problem in this run.

This packet only clears the cogenerator question from arXiv:0805.1039. It does not settle the source paper’s separate questions about Hilbert-space resolvent-integrability converses, weak stability of unitary groups, or characterizing sampling sequences  $\{t_n\}$  for weak stability.

#### SEARCH EVIDENCE

Before packaging, the lightweight run indexes were searched for arXiv:0805.1039, the title phrase, “cogenerator,” and “weak stability.” No exact prior packet was found. Local parsed sources for arXiv:0805.1039 and arXiv:2005.03821 were then inspected. A web/arXiv search for the two arXiv identifiers confirmed the paper metadata and the supporting answer source.

#### REFERENCES

- [1] Tanja Eisner, Balint Farkas, Rainer Nagel, and Andras Sereny, *Weakly and almost weakly stable  $C_0$ -semigroups*, arXiv:0805.1039, 2008. <https://arxiv.org/abs/0805.1039>
- [2] Robert E. O’Brien, *Contractions, Cogenerators, and Weak Stability*, arXiv:2005.03821, 2020. <https://arxiv.org/abs/2005.03821>